

Lecture 20 Notes: Stock Valuation, DDM, and Market Efficiency

Corporate Finance

I. Price vs. Intrinsic Value

Key Distinction

Price (P_0)	Intrinsic Value ($\hat{P}_0$)
What the market commands	True value of the stock
Easily observable (e.g., PSX database)	Based on true cash flows and true risk
Current trading price	Estimated by analysts
Market price today	What the price should be

Fundamental Principle

Price of any asset = Present Value of its expected future cash flows

Market Equilibrium

Equilibrium occurs when $P_0 = \hat{P}_0$

Condition	Action
$\hat{P}_0 > P_0$ (Undervalued)	BUY (price should rise)
$\hat{P}_0 < P_0$ (Overvalued)	SELL (price should fall)
$\hat{P}_0 = P_0$ (Fairly valued)	HOLD

Market Efficiency

- In **efficient financial markets**, equilibrium is achieved quickly.
- In less efficient markets, equilibrium may be delayed.
- Intrinsic value is ultimately an **estimate** based on perception of cash flows and risk.
- Different analysts may have different estimates (e.g., Warren Buffett).

II. Case Study: UBL (United Bank Limited)

UBL Strategy Under CEO Jawad Iqbal

Focus: Growing current accounts (0% interest accounts)

Strategy:

- Incentives for branch managers (e.g., cars for bringing in specific deposit amounts)
- Freebies: free checkbooks, free ATM usage
- Quality of service: smooth remittances, easy documentation
- Partnerships with Master Card and Visa

Why It Works:

- Cost of funds on current accounts = **0%**
- Bank invests deposits in T-bills earning **10-11%**
- Even with "freebies" costing 2-3%, the spread (profit margin) is **8% or more**

Lesson: Identifying a **good business** and **quality management** is key to financial analysis.

III. The Dividend Discount Model (DDM)

Definition

The **Dividend Discount Model (DDM)** estimates a stock's intrinsic value as the **present value of all future dividends**.

Basic DDM Formula

$$\hat{P}_0 = \frac{D_1}{(1+r_s)^1} + \frac{D_2}{(1+r_s)^2} + \frac{D_3}{(1+r_s)^3} + \dots = \sum_{t=1}^{\infty} \frac{D_t}{(1+r_s)^t}$$

If Held for One Year

$$\hat{P}_0 = \frac{D_1 + P_1}{(1+r_s)^1}$$

Suitability

- Suitable for **mature firms** that pay dividends
- Not suitable for firms that do not pay dividends or have erratic dividend patterns

IV. The Constant Growth (Gordon Growth) Model

Assumptions

1. Dividends grow at a **constant rate** (g) forever
2. The growth rate is **less than the required return** ($g < r_s$)
3. The company has a stable payout policy

The Gordon Growth Formula

$$\hat{P}_0 = \frac{D_1}{r_s - g}$$

Where:

- $D_1 = D_0 \times (1 + g)$ = Expected dividend next year
- r_s = Required return on the stock
- g = Constant growth rate
- **Condition:** $g < r_s$ (otherwise formula gives negative or infinite value)

Why Must $g < r_s$?

- In the long run, a firm cannot grow faster than the economy indefinitely.
- High growth eventually attracts competition (reducing growth).
- High growth implies higher risk (which would raise r_s).

Zero Growth Model ($g = 0$)

$$\hat{P}_0 = \frac{D}{r_s}$$

This becomes a **perpetuity** formula.

V. Determining the Growth Rate (g)

Growth comes from a company's ability to increase earnings. To grow, a company must expand capacity, which requires investment.

Company Choices for Earnings

1. **Pay Out:** Distribute all earnings as dividends → Zero growth
2. **Retain:** Keep some earnings to reinvest in assets → Positive growth

Growth Rate Formula

$$g = ROE \times b$$

Where:

- ROE = Return on Equity (Net Income / Equity)
- b = Retention Ratio = Proportion of earnings retained (reinvested)

Retention Ratio (b) = $1 - \text{Payout Ratio}$

$$\text{Payout Ratio} = \frac{\text{Dividends per Share}}{\text{Earnings per Share}}$$

VI. Required Return using CAPM

Capital Asset Pricing Model

$$r_s = r_{RF} + (r_m - r_{RF}) \times \beta$$

Where:

- r_{RF} = Risk-free rate
- $r_m - r_{RF}$ = Market risk premium
- β = Systematic risk of the stock

VII. Supernormal (Multistage) Growth Model

Some companies experience **supernormal growth** for a limited time before settling into constant long-run growth.

Examples of Supernormal Growth

- Startups with innovative products
- Companies with patents or competitive advantages
- Firms entering new markets

Example Problem

Given:

$$D_0 = \$2.00$$

Supernormal growth = 30% for 3 years

Long-run constant growth = 6%

$$r_s = 13\%$$

Step-by-Step Solution

Step 1: Calculate dividends for the supernormal period.

$$\begin{aligned}D_1 &= D_0 \times (1 + g_s) = 2.00 \times 1.30 = \$2.60 \\D_2 &= D_1 \times (1 + g_s) = 2.60 \times 1.30 = \$3.38 \\D_3 &= D_2 \times (1 + g_s) = 3.38 \times 1.30 = \$4.394\end{aligned}$$

Step 2: Calculate the first dividend of the constant growth period.

$$D_4 = D_3 \times (1 + g_c) = 4.394 \times 1.06 = \$4.658$$

Step 3: Calculate the price at the end of Year 3 (P_3).

$$P_3 = \frac{D_4}{r_s - g_c} = \frac{4.658}{0.13 - 0.06} = \frac{4.658}{0.07} = \$66.54$$

Step 4: Discount all cash flows back to Time 0.

$$\begin{aligned}\hat{P}_0 &= \frac{D_1}{(1.13)^1} + \frac{D_2}{(1.13)^2} + \frac{D_3 + P_3}{(1.13)^3} \\&= \frac{2.60}{1.13} + \frac{3.38}{1.2769} + \frac{4.394 + 66.54}{1.4429} \\&= 2.301 + 2.647 + \frac{70.934}{1.4429} \\&= 2.301 + 2.647 + 49.16 \\&= \$54.11\end{aligned}$$

Answer

$$\hat{P}_0 = \$54.11$$

VIII. Summary of DDM Models

Model	Formula
Zero Growth (Perpetuity)	$\hat{P}_0 = D/r_s$
Constant Growth (Gordon)	$\hat{P}_0 = D_1/(r_s - g)$
Supernormal Growth	Discount individual dividends + terminal value

IX. Group Project Assignment

Project Requirements

- Select a stock listed on the **Pakistan Stock Exchange (PSX)**.
- Write a report on its intrinsic value and financial model.
- Apply the valuation models discussed in class.
- Analyze the company's financial statements and growth prospects.

X. Final Syllabus

Remaining Topics

- **Chapter 9:** Stock Valuation (current topic)
- **Chapter 10:** Cost of Capital
- Handouts on **Financial Markets**
- Group Project on PSX-listed stocks

Formula Summary Sheet

Key Formulas

$$\text{Basic DDM : } \hat{P}_0 = \sum_{t=1}^{\infty} \frac{D_t}{(1 + r_s)^t} \quad (1)$$

$$\text{Constant Growth (Gordon) : } \hat{P}_0 = \frac{D_1}{r_s - g} \quad (2)$$

$$\text{Expected Dividend : } D_1 = D_0 \times (1 + g) \quad (3)$$

$$\text{Zero Growth : } \hat{P}_0 = \frac{D}{r_s} \quad (4)$$

$$\text{Growth Rate : } g = ROE \times b \quad (5)$$

$$\text{Retention Ratio : } b = 1 - \text{Payout Ratio} \quad (6)$$

$$\text{CAPM : } r_s = r_{RF} + \beta(r_m - r_{RF}) \quad (7)$$

$$\text{One-Year Holding Period : } \hat{P}_0 = \frac{D_1 + P_1}{1 + r_s} \quad (8)$$